

VAR:

x_1 = QUANTITA' PROD A in ton
 x_2 = " " B "

VINCOLI:

- $x_1 + 3x_2 \leq 10$
- $x_1 \leq 7$
- $x_1 \geq 2$

LP:

Forma Standard:

$$\begin{aligned} \max z: & x_1 + 3x_2 \\ & x_1 + 3x_2 \leq 10 \\ & x_1 \leq 7 \\ & x_1 \geq 2 \end{aligned}$$

$$\begin{aligned} \min(-z): & -x_1 - 3x_2 \\ & x_1 + 3x_2 + x_3 = 10 \\ & x_1 + x_4 = 7 \\ & x_1 - x_5 = 2 \end{aligned}$$

0	-1	-3	0	0	0
10	1	3	1	0	0
7	1	0	0	1	0
2	1	0	0	0	-1

Introduce x_i^2

MINIMIZE

$$z = \sum_{i=1}^n x_i^2 = x_1^2$$

-7	-2	-1	0	0	0	1	0
x_3	10	1	3	1	0	0	0

$$B = \{A_3, A_4, A_6\}$$

$$x = (0, 0, 10, 7, 0, 2)$$

$x_4 =$	7	1	0	0	1	0	0	$\lambda = (0, 0)$
$x_1 =$	2	(1)	0	0	0	-1	1	

$$\theta_{\max} = \min_{i: y_{i1} > 0} \frac{y_{i0}}{y_{i1}} = \min \{10, 7, 2\} = 2$$

$-z$	0	0	0	0	0	0	1	$B = \{A_3, A_4, A_1\}$
$x_3 =$	8	0	3	1	0	1	-1	$x = (2, 0, 8, 5, 0, 0)$
$x_4 =$	5	0	0	0	1	1	-1	$\beta = (2, 0)$
$x_1 =$	2	1	0	0	0	-1	1	new reduced cost

	0	-1	-3	0	0	0	0
	8	0	3	1	0	1	-1
	5	0	0	0	1	1	-1
	2	1	0	0	0	-1	1

$-z$	2	0	-3	0	0	-1	1	$B = \{A_3, A_4, A_1\}$
$x_3 =$	8	0	(3)	1	0	1	-1	$x = (2, 0, 8, 5, 0, 0)$
$x_4 =$	5	0	0	0	1	1	-1	
$x_1 =$	2	1	0	0	0	-1	1	

REGOLA DI BLAND: $\sum = 2$

$$\theta_{\max} = \min_{i: y_{i2} > 0} \frac{y_{i9}}{y_{i2}} = \left\{ \frac{8}{3} \right\} = \frac{8}{3}$$

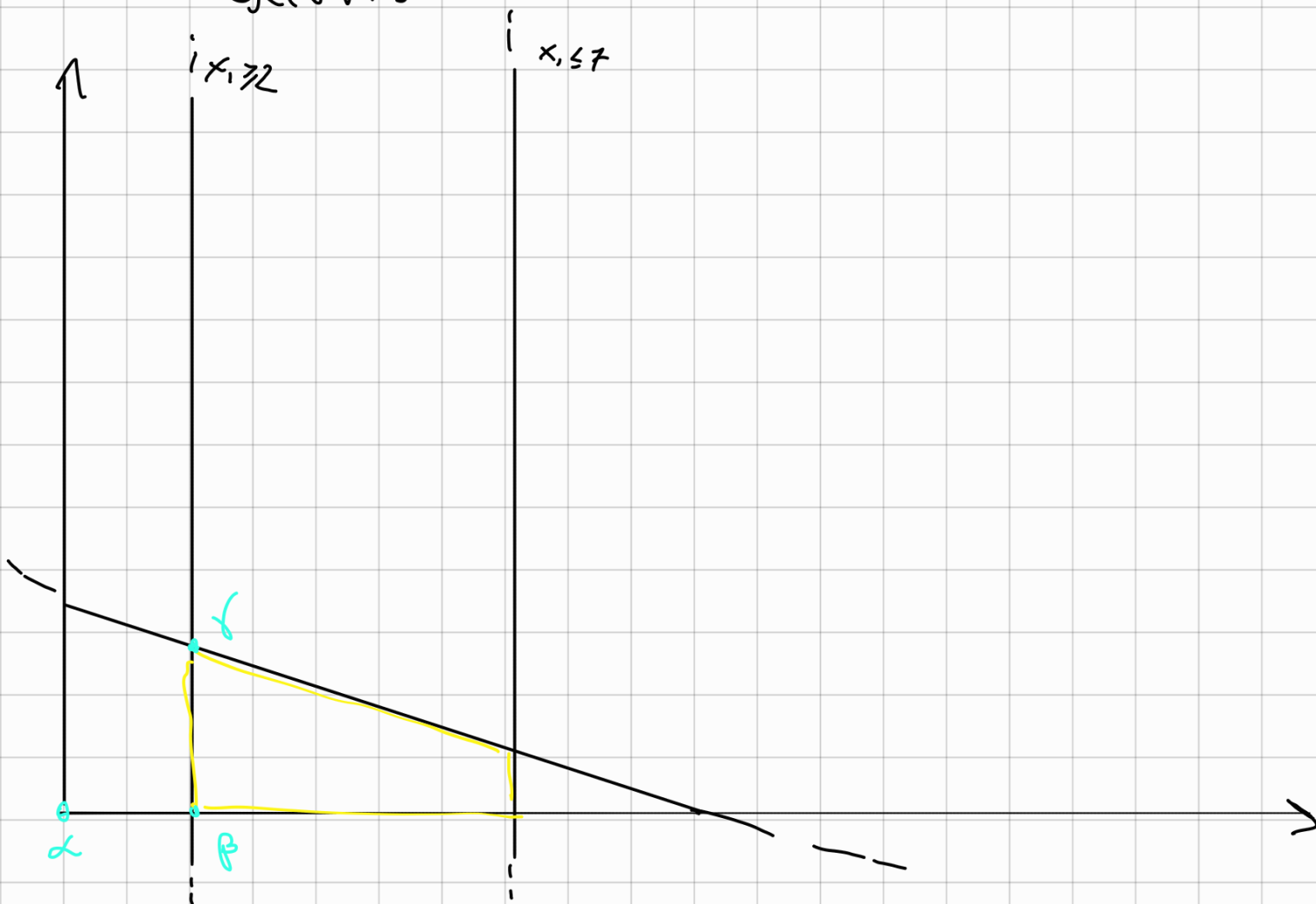
Z	10	0	0	1	0	0	0
$x_2 = \frac{8}{3}$	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$	$-\frac{1}{3}$	
$x_4 = 5$	0	0	0	1	1	-1	
$x_1 = 2$	1	0	0	0	-1	1	

$$B = \{A_2, A_4, A_1\}$$

$$x = (2, \frac{8}{3}, 0, 5, 0, 0)$$

$$x = (2, \frac{8}{3})$$

OTTAVA!



$$\min(z) = x_1 + 2x_2$$

$$\min(-E): -x_1 - 5x_2$$

$$x_1 + 3x_2 + x_3 = 10$$

$$x_1 + x_4 = 7$$

$$x_1 - x_5 = 2$$

DUAL:

$$\max W: 10\pi_1 + 7\pi_2 + 2\pi_3$$

$$\pi_1 + \pi_2 + \pi_3 \leq -1$$

$$3\pi_1 \leq -3$$

$$\pi_1 \leq 0$$

$$\pi_2 \leq 0$$

$$-\pi_3 \leq 0$$

$$\pi_1, \pi_2, \pi_3 \geq 0$$

$$\text{sol. opt. DUAL } \pi_3 = c_3 - \bar{c}_3$$

$$c_3 = (0, 0, 0), \quad \bar{c}_3 = (1, 0, 0)$$

$$\pi_3 = (-1, 0, 0)$$

Z	10	0	0	1	0	0
$x_2 = \frac{8}{3}$		0	1	$\frac{1}{3}$	0	$\frac{1}{3}$
$x_4 = 5$		0	0	0	1	1
$x_1 = 2$		1	0	0	0	-1

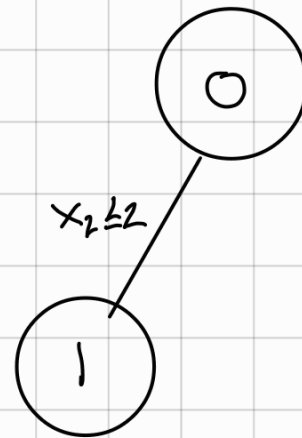
x_2 BRANCH & BOUND

$$x_2 \leq \left\lfloor \frac{8}{2} \right\rfloor \rightarrow x_2 \leq 2$$

$$UB = \lfloor 10 \rfloor = 10$$

FORMA STANDARD:

$$x_2 + x_6 = 2$$



10	0	0	1	0	0	0
$\frac{8}{3}$	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0
5	0	0	0	1	1	0
2	1	0	0	0	-1	0
2	0	1	0	0	0	1

$Z =$	10	0	0	1	0	0	0
$x_5 =$	$\frac{8}{3}$	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0
$x_4 =$	5	0	0	0	1	1	0
$x_1 =$	2	1	0	0	0	-1	0
$x_6 =$	$-\frac{2}{3}$	0	0	$-\frac{1}{3}$	0	$-\frac{1}{3}$	1

SIMP. TABLE

$$\hat{i} = 4$$

Pivot:

$$\max_{S: y_{45} \leq 0} \frac{y_{05}}{y_{45}} = \frac{y_{05}}{y_{45}} = \max \left\{ -3, 0 \right\} = 0$$

$Z =$	10	0	0	1	0	0	0
$x_2 =$	2	0	1	0	0	0	1
$x_4 =$	3	0	0	-1	1	0	3
$x_1 =$	4	1	0	1	0	0	-3
$x_3 =$	2	0	0	1	0	1	-3

$$B = \{A_2, A_4, A_1, A_3\}$$

$$x = (4, 2, 0, 3, 2, 0)$$

$$s = (4, 2)$$